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High-bandwidth GaN Substrate Single-Pixel Blue Micro-LED Toward 10 Gbps Visible Light Communication

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Abstract: A record data rate of 10.009 Gbps at a distance of 0.55 m based on a single c-plane freestanding GaN micro-LED was achieved by using OFDM modulation and a bit-loading algorithm. © 2024 The Author(s)

1. Introduction

The increasing data volume has led to an increasing demand for high-speed communication. In the future, there will be more demands for 6G communication with high capacity [1,2]. Therefore, visible light communication (VLC) is proposed as an important technology for 6G communication due to its ability to meet the needs of communication development. Micro-LEDs have been regarded as one of the most promising transmitters for VLC systems because of the characteristics of high modulation bandwidth. Recently, significant progresses have been achieved for the application of micro-LED in VLC [3-9].

Recently, series-connected micro-LEDs have been demonstrated to achieve high-speed VLC over 10 Gbps [3-4]. In [3], a data rate of 11.74 Gbps VLC system was achieved by using a 3 × 3 series-biased blue micro-LED array and orthogonal frequency division multiplex (OFDM) modulation. The data rate has reached 10.23 Gbps based on a 3 × 3 series-biased violet micro-LED array [4]. The current data rates of micro-LEDs have exceeded 10 Gbps, but the way of breaking 10 Gbps is mainly based on the series-connected micro-LEDs. To date, there is still a lack of research achieving over 10 Gbps data rate by a single micro-LED. A VLC system based on a violet micro-LED was reported to achieve a data rate of 7.91 Gbps at a free space of 0.08 m by using OFDM modulation and an adaptive bit-loading algorithm [5]. A data rate of 8.75 Gbps was achieved by using a high-bandwidth single blue wetting layer micro-LED [6]. The above studies fully demonstrated the significant advantages of micro-LED for VLC. However, the above studies have not achieved the VLC system with a data rate of over 10 Gbps by using a single micro-LED. In addition, some research has shown that GaN-substrate micro-LEDs have lower threading dislocation density (TDD) and better heat dissipation, which provide GaN-substrate micro-LEDs with higher LOP and modulation bandwidth [7]. Thus using GaN-substrate micro-LED as a VLC transmitter will be able to increase the data rate [8-10].

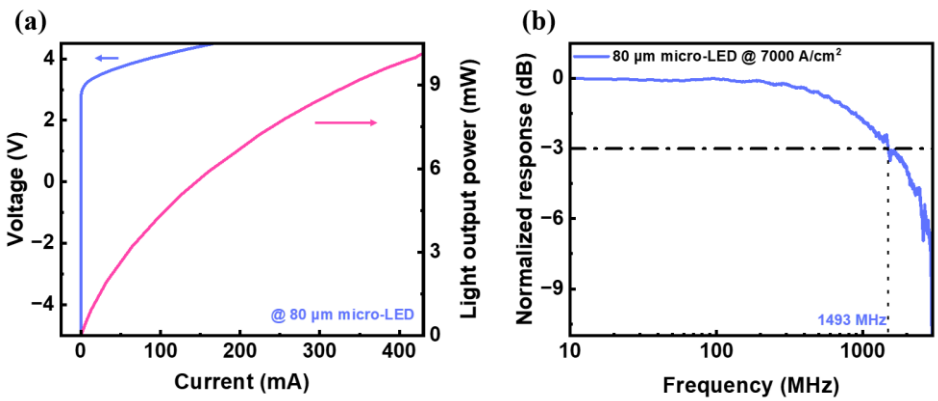

Fig. 1. (a) V-I and P-I characteristics of an 80 μm blue micro-LED. (b) The frequency response of the blue micro-LED at the current density of 7000 A/cm².

In this paper, we proposed a VLC system based on a c-plane freestanding blue GaN-substrate micro-LED. We investigated the optoelectronic characteristics and modulation bandwidth performance of the micro-LED with a size of 80 μm and demonstrated its high modulation bandwidth and high LOP. Using this micro-LED as a transmitter and with a combination of OFDM modulation and bit-loading algorithms, a data rate of up to 10.009 Gbps was achieved at a distance of 0.55 m. To the best of our knowledge, this is the highest data rate of VLC systems based on single-pixel micro-LED.

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2. Devices Fabrication and Experimental setup

We investigated the optoelectronic characteristics and modulation bandwidth performance of 80 μm c-plane freestanding blue GaN-substrate micro-LED. Fig.1(a) shows the characteristics of voltage versus current (V-I) and light output power versus current (P-I). The light output power of the micro-LED was measured by an integrating sphere (IS-25-LED). It can be seen that the LOP of the micro-LED can reach 10.125 mW when the current is increased to 430 mA, which shows the high optical power characteristics of the micro-LED. We measured the frequency response of the device at the current density of 7000 A/cm² by using a network analyzer (VNA, PicoVNA 106) as shown in Fig.1(b). The -3 dB bandwidth is 1493 MHz, which is beneficial to the transmission of high-frequency signals to achieve higher rates.

The schematic diagram of the experimental setup of the OFDM VLC system based on the blue micro-LED is shown in Fig.2(a). Pre-equalization technology and the bit-loading algorithm were used to improve communication performance. Pre-equalization was utilized in the frequency domain to expand the available signal bandwidth. The bit-loading algorithm can improve the spectral efficiency of the VLC system. Fig.2(b) shows the microscope image of the micro-LED with a size of 80 μm. Fig.2(c) shows the experimental setup photograph of the 0.55 m free-space VLC system. At the transmitter, the OFDM data generated by the MATLAB program was uploaded to an arbitrary waveform generator (AWG, Keysight M8190A, 12 GS/s) from the transmitter (Tx) module with the AWG sampling rate of 8 GS/s. A 38 dB electrical amplifier (Mini-Circuits, ZHL-42W+, 4.2 GHz) was used to amplify the output signal and then combined with a DC signal via a bias-Tee (Pasternack PE1615, 6 GHz) to drive the micro-LED. At the receiver, the optical signal emitted from the micro-LED after modulation was collimated and focused into a 1.4 GHz PIN photodiode (PIN, Femto HSPR-X-I-1G4-SI, 1.4 GHz) to convert the optical signal into an electrical signal. The high-speed digital signal analyzer (DSA, Agilent DSA90604A Infiniium 20 GS/s) acquired and recorded the output signal of the receiver, and the final output signal was demodulated using MATLAB [11].

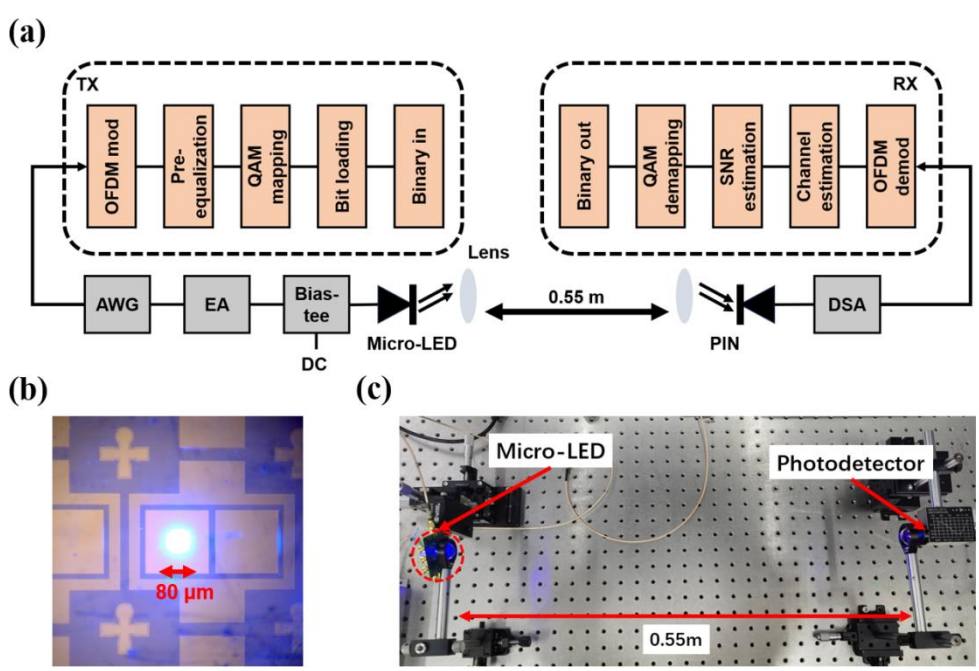

Fig. 2. (a) The schematic diagram of the experimental setup for the VLC system. (b) The microscope images of the blue micro-LED. (c) The experimental setup photograph of the 0.55 m free-space VLC system based on the micro-LED.

3. Results and discussion

The communication performance of an 80 μm blue micro-LED was evaluated at a distance of 0.55 m. We first determined the optimal driving current and signal peak-to-peak voltage (V_{PP}) for the device are 450 mA and 0.7 V, respectively. Next, we estimated the SNR of each subcarrier by transmitting the OFDM quadrature phase shift keying (QPSK) signal as a training signal and calculated the bit number as shown in Fig.3(a). The average spectral efficiency could be calculated from the bit loading result, which can reach 5.0045 bit/s/Hz. The average SNR was 19.6 in the range of 0-2 GHz. Fig.3(b) shows the power spectrum, which is relatively flat in the 2 GHz signal bandwidth through pre-equalization technology. The data rate is calculated by the product of spectral efficiency and signal bandwidth

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[12]. The curve of BER versus data rate is shown in Fig.3(c). It can be seen that the data rate is up to 10.009 Gbps under the forward error correction (FEC) threshold of 3.8×10^{-3}.

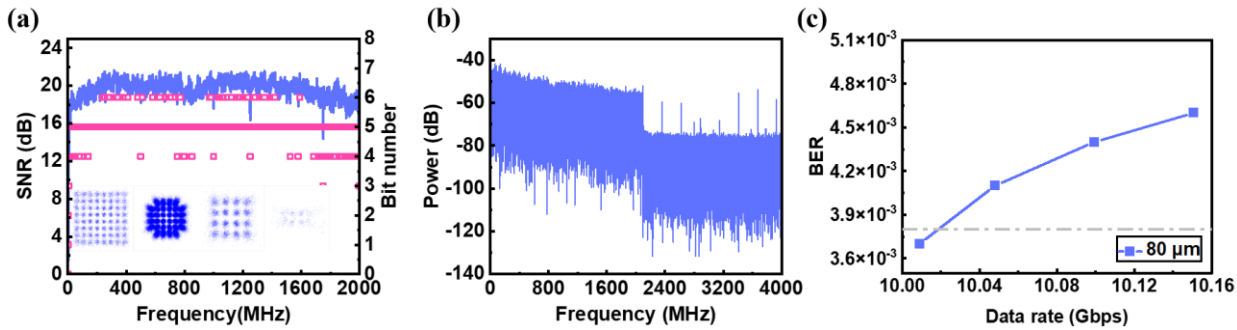

Fig. 3. (a) SNR distribution, bit distribution, and constellation diagram of 80 μm micro-LED. (b) The OFDM signal power spectrum of 80 μm micro-LED. (c) BER versus different data rates of the VLC system.

4. Conclusions

In this paper, we experimentally demonstrated a high-speed VLC system based on the c-plane freestanding blue GaN-substrate micro-LED and investigated the optoelectronic characteristics and modulation bandwidth performance of the device. The micro-LED had an optical power of 10 mW and a modulation bandwidth of 1493 MHz. An 80 μm blue single micro-LED was used as a transmitter in the VLC system and a data rate of 10.009 Gbps was achieved below the FEC limit of 3.8×10^{-3} by using OFDM modulation and a bit-loading algorithm with a free space transmission distance of 0.55 m.

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